

2.8.1

EE25BTECH11022 - sankeerthan

Question: If $(-4, 3)$ and $(4, 3)$ are two vertices of an equilateral triangle. Find the coordinates of the third vertex, given that the origin lies in the interior of the triangle.

solution: Let

$$A = \begin{pmatrix} -4 \\ 3 \end{pmatrix}, B = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

Consider rotating A about B by 60° . Let the rotation matrix be:

$$R_{60^\circ} = \begin{pmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \quad (1)$$

Translate B to the origin:

$$A' = A - B = \begin{pmatrix} -4 - 4 \\ 3 - 3 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (2)$$

Translate back:

$$C = B + R_{60^\circ}(A) \quad (3)$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (4)$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \times (-8) + \left(-\frac{\sqrt{3}}{2}\right) \times 0 \\ \frac{\sqrt{3}}{2} \times (-8) + \frac{1}{2} \times 0 \end{pmatrix} \quad (5)$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} -4 \\ -4\sqrt{3} \end{pmatrix} \quad (6)$$

$$= \begin{pmatrix} 4 + (-4) \\ 3 + (-4\sqrt{3}) \end{pmatrix} \quad (7)$$

$$= \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \quad (8)$$

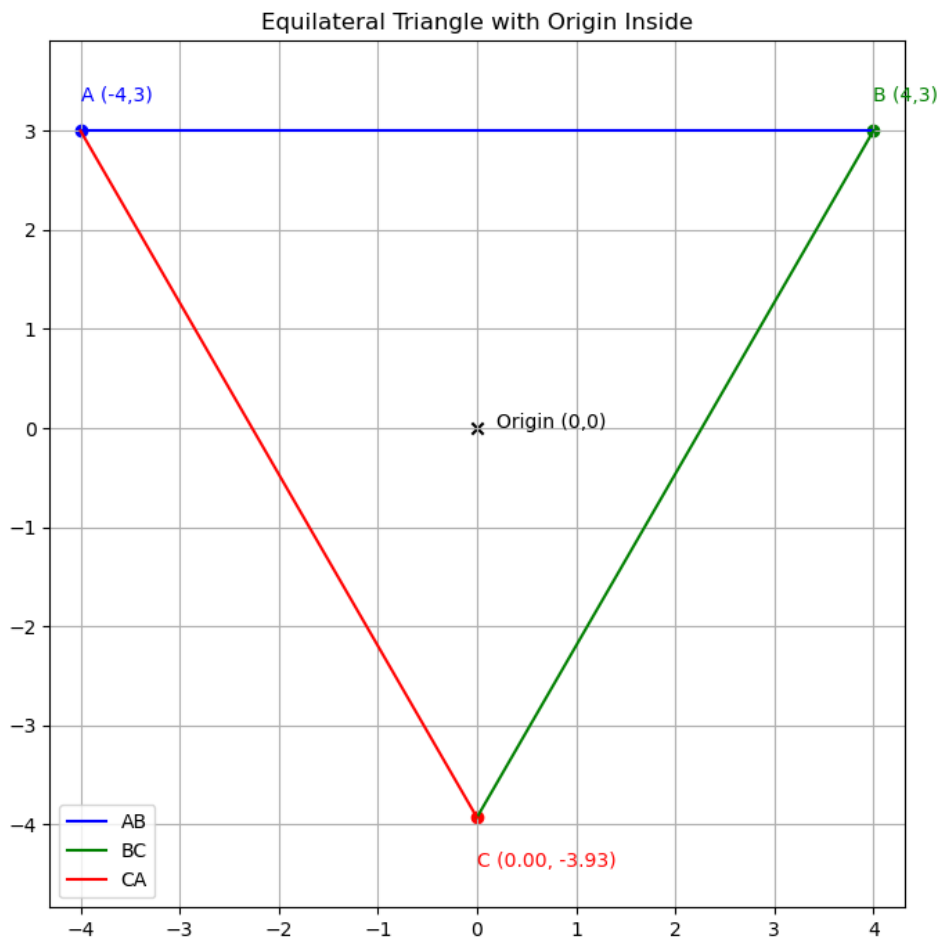


Fig. 0.1